

Boolean Function Analysis

Given: $F(A, B, C) = \Sigma m(2, 4, 6, 7)$

1. Truth Table

A	B	C	Decimal	F
0	0	0	0	0
0	0	1	1	0
0	1	0	2	1
0	1	1	3	0
1	0	0	4	1
1	0	1	5	0
1	1	0	6	1
1	1	1	7	1

Note: Output $F = 1$ for minterms 2, 4, 6, 7.

2. Canonical SOP (Sum of Products)

Write product terms for rows where $F = 1$:

- $m_2: \overline{A} B \overline{C}$
- $m_4: A \overline{B} \overline{C}$
- $m_6: A B \overline{C}$
- $m_7: A B C$

$$F_{\text{SOP}} = \overline{A} B \overline{C} + A \overline{B} \overline{C} + A B \overline{C} + A B C$$

3. Canonical POS (Product of Sums)

$F = 0$ for minterms 0, 1, 3, 5 $\rightarrow$ maxterms M_0, M_1, M_3, M_5 :

- $M_0: A + B + C$
- $M_1: A + B + \overline{C}$
- $M_3: A + \overline{B} + \overline{C}$
- $M_5: \overline{A} + B + \overline{C}$

$$F_{\text{POS}} = (A + B + C)(A + B + \overline{C})(A + \overline{B} + \overline{C})(\overline{A} + B + \overline{C})$$

4. Simplified SOP (Using Boolean Laws)

Start with canonical SOP:

$$F = \overline{A} B \overline{C} + A \overline{B} \overline{C} + A B \overline{C} + A B C$$

Step-by-step simplification:

1. Group terms with A :

$$F = \overline{A} B \overline{C} + A(\overline{B} \overline{C} + B \overline{C} + B C)$$

2. Apply **Distributive + Inverse Law**:

$$\overline{B} \overline{C} + B \overline{C} = (\overline{B} + B)\overline{C} = 1 \cdot \overline{C} = \overline{C}$$

3. Now:

$$F = \overline{A} B \overline{C} + A(\overline{C} + B C)$$

4. Simplify $\overline{C} + B C$ using **Distributive Law**:

$$\overline{C} + B C = (\overline{C} + B)(\overline{C} + C) = (\overline{C} + B)(1) = \overline{C} + B$$

5. Distribute A :

$$F = \overline{A} B \overline{C} + A \overline{C} + A B$$

6. Combine terms: $F = A B + A \overline{C} + B \overline{C}$

✓ **Final Simplified SOP:**

$$\boxed{F = A B + A \overline{C} + B \overline{C}}$$

Laws used: Distributive, Inverse, Identity, Redundancy.

5. Simplified POS (Algebraic Conversion)

Start from simplified SOP:

$$F = A B + A \overline{C} + B \overline{C}$$

1. Find $\overline{F}$ using **De Morgan's Law**:

$$\overline{F} = \overline{A B + A \overline{C} + B \overline{C}} = (\overline{A} + \overline{B})(\overline{A} + C)(\overline{B} + C)$$

2. Then $F = \overline{\overline{F}}$. But instead, simplify $\overline{F}$ and re-complement:

$$\overline{F} = \overline{A} \overline{B} + \overline{A} C + \overline{B} C$$

3. Apply **De Morgan’s Law** again: $F = \overline{\overline{A} \overline{B} + \overline{A} C + \overline{B} C} = (A + B)(A + \overline{C})(B + \overline{C})$

✔ **Final Simplified POS:**

$$F = (A + B)(A + \overline{C})(B + \overline{C})$$

✔ Verified with truth table — matches minterms 2, 4, 6, 7.

Summary

Form	Expression
Canonical SOP	$\overline{A} B \overline{C} + A \overline{B} \overline{C} + A B \overline{C} + A B C$
Simplified SOP	$A B + A \overline{C} + B \overline{C}$
Canonical POS	$(A + B + C)(A + B + \overline{C})(A + \overline{B} + \overline{C})(\overline{A} + B + \overline{C})$
Simplified POS	$(A + B)(A + \overline{C})(B + \overline{C})$

♦ All derivations use **Boolean algebra laws only** — no K-maps, no circuits.